Jacob Wilson

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EDUCATION

University of Southern Mississippi — Ph.D. in Computer Science — Fall 2026–Present
First-year doctoral student

Mississippi State University — B.S. in Computer Science — May 2026
Mathematics minor · Shackouls Honors College · GPA: 3.47 overall / 3.72 major

RESEARCH

Quasi-Novelty / Reconciliation — 2026–Present
Continual reinforcement learning, information rewards, and learning dynamics
Originated as undergraduate research with Dr. Jingdao Chen, Mississippi State University.

- Developed recurrent reinforcement learning agents from random weights to study sustained online learning, using intrinsic information rewards without pretraining, demonstrations, or command-correctness labels. Investigated recurrent memory and immediate versus episode-duration credit.
- Analyzed the directions and magnitudes of gradients and actual parameter changes, combining update geometry, entropy measurements, and controlled interventions to distinguish useful adaptation from optimizer suppression, repetitive behavior, and weak observation dependence.
- Identified a transient observation-conditioned routine producing 36 execution-verified pointer-following chains across 26 distinct command sequences. Audited action dependencies and seven unsuccessful fresh replications from random initialization to characterize acquisition bottlenecks and reproducibility limits.
- Investigated loss of behavioral access versus erasure: attenuating carried adaptive parameter offsets restored 24/24 responses from 0/24 on selected recorded contexts without retraining persistent weights. Used this result to motivate separate prospective tests of acquisition, retention, recovery, and transfer.

Manuscript: Quasi-Novelty Revisited. Working manuscript, October 2026.

EXPERIENCE

Teaching Assistant — University of Southern Mississippi — Fall 2026–Present

- Tutor students weekly and grade coursework for Computer Networks and Computer Systems.

Software Engineering Consultant — HUMO, Starkville, MS — Feb–Oct 2023

- Developed motion-tracking software with faculty and graduate students for golf-swing analysis, applying sensor fusion and signal processing to real-time pose estimation from 9DOF and 6DOF IMUs.
- Built visualization tools in C#, Unity, and C; analyzed sensor data in Python; developed a Swift/iOS testing app and collaborated on a C++/Qt prototype.

ENTREPRENEURSHIP

Founder, Aebyss Technologies — 2023–Present

- Developed a Swift proof-of-concept study app for an AI-based EdTech startup; won second place and a $1,000 grant at the 2023 MSU Startup Summit. Currently in Stage 3 of MSU’s VentureCatalyst program.

TECHNICAL SKILLS

Languages: Python, C#, C++, C, Swift, Rust, JavaScript
Tools: PyTorch, Git, Unity, Qt, Xcode, React
Research: Reinforcement learning, recurrent networks, information rewards, linear algebra, numerical analysis